VENKATA PAVAN KANDULA

DevOps Engineer II | Kubernetes & Cloud Infrastructure Specialist
Hyderabad, Telangana, India | +91-8341950542 | pavankandula62@gmail.com | Portfolio | LinkedIn | GitHub

PROFESSIONAL SUMMARY

Results-driven DevOps Engineer with 4.2 years of hands-on experience scaling and securing enterprise Kubernetes infrastructure on AWS EKS and managing high-throughput datastore (RDS). Proven track record at TCS managing 20+ microservices across multi-environment promotion chains, with deep expertise in Terraform IaC, GitHub Actions CI/CD, observability stacks (Prometheus, Grafana), Kubernetes autoscaling (HPA/VPA), disaster recovery planning, and security compliance frameworks. Delivered a 40% reduction in release cycle time, 30% AWS cost savings through right-sizing and Reserved Instance optimization, and eliminated 8–10 hours of weekly manual toil through Python automation.

CORE TECHNICAL SKILLS

Kubernetes / EKS: EKS, AKS cluster design, RBAC, network policies, Helm, pod security standards, ArgoCD GitOps, HPA/VPA autoscaling, cluster autoscaler
CI/CD: GitHub Actions (primary), GitLab CI/CD, Jenkins, AWS CodePipeline, SonarQube, Artifactory, Nexus Repository, ArgoCD — multi-env pipelines Dev→QA→UAT→Prod
Cloud (AWS): EC2, EKS, S3, RDS, Lambda, IAM, VPC, CloudWatch, Secrets Manager, AWS Cost Explorer, Reserved Instances
Infrastructure as Code: Terraform (modules, workspaces, remote state), Helm chart authoring, AWS CloudFormation, GitOps workflows
Disaster Recovery & Backup: RDS automated backups, multi-AZ failover, DR runbooks, RTO/RPO planning, cross-region S3 replication
Security & Compliance: Zero-trust architecture, container image scanning (Trivy), TLS/mTLS, RBAC, IAM least-privilege, secrets management.
Cost Optimization: AWS cost governance, right-sizing, Reserved Instance planning, Spot Instance strategy, resource tagging, Cost Explorer dashboards.
Datastores: AWS RDS, PostgreSQL.
Monitoring & Observability: Grafana, Prometheus, CloudWatch — real-time dashboards & alerting.
Collaboration & Agile: Jira, Agile/Scrum.
Scripting & Automation: Python (self-healing systems, predictive scaling, automation), Groovy (Jenkins shared libraries)
Networking & Security: VPC design, ALB/NLB, DNS, TLS/mTLS, OPA/Gatekeeper, zero-trust architecture, container image scanning

PROFESSIONAL EXPERIENCE

AWS DevOps Engineer | Tata Consultancy Services (TCS) — Hyderabad, India | *Mar 2022 – Present (4 Years)*

- **Kubernetes Autoscaling (HPA/VPA):** Designed and deployed Horizontal Pod Autoscaler (HPA) and Vertical Pod Autoscaler (VPA) policies across 20+ microservices on AWS EKS; combined with Cluster Autoscaler for node-level scaling — ensuring SLA adherence during traffic spikes and reducing over-provisioning by ~25%.
- **Kubernetes Cluster Management :** Provisioned and managed production-grade Kubernetes clusters on AWS EKS; enforced RBAC, network policies and pod security standards; performed cluster upgrades with zero downtime.
- **ArgoCD GitOps Deployments:** Implemented and maintained ArgoCD-based GitOps workflows for consistent, automated application deployments across Dev→QA→UAT→Prod; ensured single source of truth for all Kubernetes manifests and Helm charts, reducing config drift and rollback time.
- **AWS Cost Optimization (30% Savings):** Led cloud cost optimization initiative using AWS Cost Explorer and Trusted Advisor — identified and right-sized over-provisioned EC2 and RDS instances, implemented Reserved Instance purchasing strategy, and introduced Spot Instances for non-critical workloads; achieved ~30% reduction in monthly AWS spend.
- **Disaster Recovery & Backup Strategy:** Architected and implemented DR strategy across environments — configured RDS automated backups with point-in-time recovery, r, and cross-region S3 replication for critical artifacts; defined RTO/RPO targets and authored DR runbooks for P1 incident scenarios.
- **Security Compliance & Hardening:** Enforced security compliance across Kubernetes clusters using OPA/Gatekeeper policy-as-code for admission control; implemented container image scanning with Trivy in CI pipelines; applied IAM least-privilege principles, rotated secrets via AWS Secrets Manager, and aligned infrastructure controls with SOC 2 requirements.
- **Datastore Management (Elasticsearch, RDS, DocumentDB):** Administered Elasticsearch clusters for ELK observability stack; managed AWS RDS (PostgreSQL,; performed performance tuning, backup policies, and scaling operations to support high-throughput workloads.
- **Terraform Infrastructure as Code:** Authored modular Terraform configurations for all AWS infrastructure (VPC, EKS, RDS, S3, IAM, Lambda, API Gateway); used workspaces and remote state for environment isolation; maintained GitOps-aligned IaC repositories enabling repeatable, auditable provisioning.
- **GitHub Actions CI/CD Pipelines:** Designed and maintained end-to-end CI/CD pipelines using GitHub Actions, GitLab CI/CD, and Jenkins; integrated automated build, test, security scanning (Trivy, SAST), and deployment stages — achieving a 40% reduction in release cycle time; on-boarded multiple applications onto the DevOps toolchain.

- **Monitoring & Observability (Datadog / Grafana / ELK):** Deployed enterprise observability stack using Prometheus + Grafana; built real-time dashboards for cluster health, pipeline metrics, and datastore performance; implemented intelligent alerting that significantly reduced MTTR.
- **Python Automation:** Developed Python scripts eliminating 8–10 hours of weekly manual toil for infrastructure health checks, artifact management, and operational reporting; built self-healing Kubernetes operators for automatic workload recovery and incident remediation.
- **Engineering Team Support & On-Call:** Provided 24/7 on-call support; resolved P1/P2 incidents within defined SLA; conducted RCA for critical failures; mentored junior engineers on Kubernetes, Terraform, and CI/CD best practices; produced architecture diagrams and operational runbooks.

PROJECTS

Motor Insurance Platform — AWS Cloud & DevOps Engineering | TCS — Hyderabad, India | Ongoing

A cloud-native Motor Insurance application deployed on AWS, enabling customers to purchase policies, renew coverage, and manage claims online. The platform is built on a microservices architecture and hosted on AWS EKS, leveraging core AWS services including EC2, EKS, ECR, RDS, S3, IAM, Route53, and CloudWatch. Infrastructure is provisioned using Terraform, with Jenkins driving CI/CD automation for all deployment pipelines.

Role & Responsibilities (DevOps Engineer):

- Managed AWS infrastructure and EKS clusters to ensure high availability and reliability of the insurance platform across all environments.
- Provisioned and maintained cloud resources using Terraform, including VPC, EKS, RDS, S3, IAM roles and policies, Route53 DNS, and ECR repositories.
- Built and deployed Dockerized microservices to AWS EKS through Jenkins CI/CD pipelines, enabling automated build, test, and deployment workflows.
- Monitored application and infrastructure health using CloudWatch, Prometheus, and Grafana; configured alerts and dashboards for proactive incident detection.
- Troubleshot production issues, performed application deployments and rollbacks, and ensured minimal downtime during release windows.
- Managed secrets and access controls using AWS IAM and Secrets Manager, enforcing least-privilege principles across all services.
- Developed Python automation scripts to reduce manual operational effort, streamline infrastructure health checks, and improve overall team efficiency.

Tech Stack: AWS (EC2, EKS, ECR, RDS, S3, IAM, Route53, CloudWatch), Terraform, Jenkins, Docker, Kubernetes, Prometheus, Grafana, Python Scripting

KEY ACHIEVEMENTS

- 40% reduction in release cycle time through end-to-end CI/CD pipeline optimization across GitHub Actions and Jenkins.
- ~30% AWS cost savings achieved via right-sizing, Reserved Instance strategy, and Spot Instance adoption — identified through Cost Explorer and Trusted Advisor analysis.
- Eliminated 8–10 hours of weekly manual toil via Python automation for infrastructure health checks, artifact management, and reporting.
- HPA/VPA + Cluster Autoscaler across 20+ microservices — ~25% over-provisioning reduction; RDS PITR, and cross-region S3 replication.
- Security hardening via OPA/Gatekeeper, Trivy, IAM least-privilege & SOC 2 controls; enterprise observability (Prometheus + Grafana) driving significant MTTR reduction.

CERTIFICATIONS

AWS Certified DevOps Engineer - Professional

Issued: April 18, 2026 | Valid Through: April 18, 2029

Validation Number: 45d35f868ff14ecfb34d5ac2703ac088

Verify: https://www.credly.com/badges/8d53ab72-a454-4977-829e-0dfb6bee6a44/public_url

EDUCATION

Bachelor of Technology (B.Tech) — Mechanical Engineering

Aditya Engineering College, JNTU Kakinada | Graduated October 2020